TIC 341290373

Eclipsing binary candidate dossier - Star Vetter

THIN MARGIN

```
BLS EPHEMERIS
  Period        1.874806 d
  Epoch (t0)    1598.3938 BTJD
  Duration      0.0840 d
  Depth         2.496 %  (24,956 ppm)
  SNR           209.9

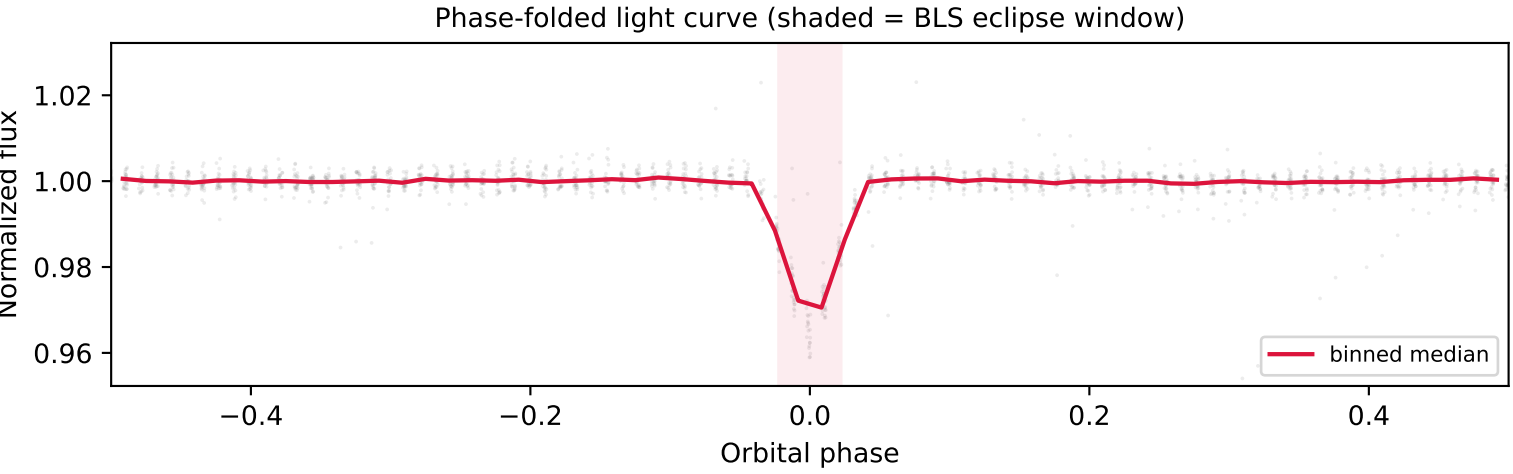
INDEPENDENT CHECKS (f3i gates)
  Eclipses observed      22  (gate >= 3)
  Robust depth ratio     1.04  (gate >= 0.50)
  Robust depth sigma     122.7  (gate >= 7.0)
  Odd/even z-score       2.42  (consistent)
  Secondary eclipse      not detected
```

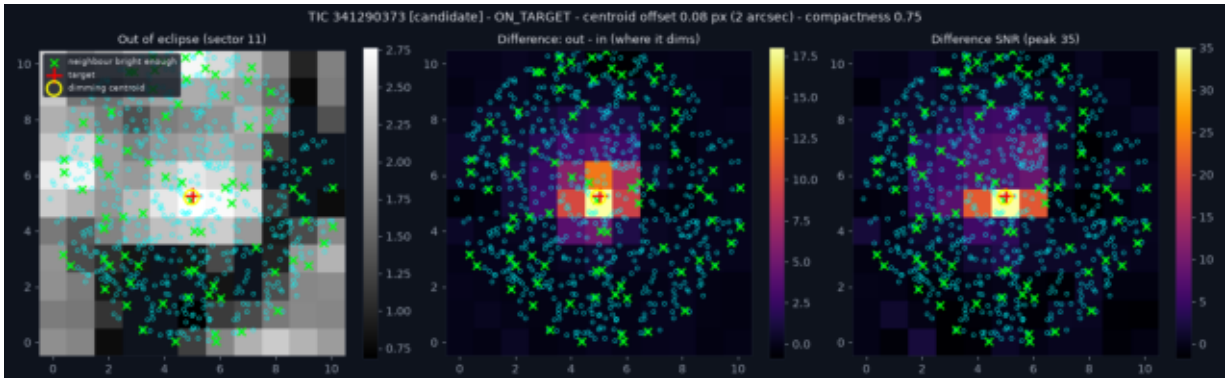
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CATALOG CROSS-MATCH (novelty)

  VSX                                  no match
  ASAS-SN                              no match
  Gaia eclipsing-binary table          no match
  Gaia variable classification         no match
  TESS EB catalog (Prsa+2022)          no match
  ExoFOP TOI list                      no match

  Verdict: NOVEL
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PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)
  Sector 11    light-curve dip in this sector: 76.4 sigma    t0 refined by 0.0050 phase
  Difference-image peak SNR   35.04   (gate >= 5.0)      Compactness   0.75   (gate >= 0.50)
  Dimming centroid offset     0.08 px (2 arcsec)   (gate <= 1.0 px, i.e. on the target)
  Nearest bright-enough neighbour in cutout: TIC 341290377
    0.11 px away, dTmag -0.79 - farther from the dimming than the target
```





Caveat (THIN MARGIN): The ON_TARGET call rests on a neighbour-competition margin of just 0.029 px (dimming centroid 0.083 px from target vs 0.112 px from the nearest bright-enough neighbour) - inside diag_pixel.py's own stated centroid systematic of 0.10-0.3 px. TESS's resolution cannot confidently separate this star from that neighbour.